

Final Project: Analyzing a Data Breach - Tasks and Instructions

In this project, you'll first select an appropriate case study for evaluation. You'll then use the standard case study analysis template to document your observations. The template being referred to is the same case study analysis template that you've used throughout the course to work through each case study.

Your tasks will include the following:

- **Task 1:** Find a suitable cybersecurity case study that you would like to evaluate and write a concise introduction (in up to 150 words) to set the context for analysis and engagement.
(Note: The selected case study should not be from the ones discussed in the course.)
- **Task 2:** Provide two credible citations that reference your selected case study.
- **Task 3:** Use the standard case study analysis template to document your observations.

Read your chosen case study thoroughly, document your response against each section, and ensure that all information is concise and relevant to enhance understanding and application. You are required to type your responses in the text boxes provided below each task. You should plan to complete the project in about 90 minutes.

Step-by-step instructions

Refresh your knowledge

Before starting on the project or while doing the project, you can refresh your knowledge by referring to the following lessons:

- Module 1, Lesson 1: Analyzing Case Studies
- Module 1, Lesson 2: Phishing Scams
- Module 2, Lesson 1: Point of Sale (PoS) Breaches
- Module 2, Lesson 2: Insider Breaches
- Module 3, Lesson 1: AI-Related Breaches
- Module 3, Lesson 2: Ransomware Breaches
- Module 4, Lesson 1: Incident Response
- Module 4, Lesson 2: Digital Forensics

Project scenario

Imagine being employed as a cybersecurity professional at a consulting firm. The organization recently suffered a data breach, compromising sensitive customer information, including financial and personal data. The executive team has assigned your department the responsibility of examining a case study concerning a similar incident at another organization, with the objective of deriving insights and proposing strategies to avert future breaches. Your assignment is to conduct a structured analysis of the case study using the template provided by the organization to formulate actionable recommendations.

Completing project tasks

Submitting a complete assignment by adhering to the instructions will facilitate a thorough review.

For successful project submission:

- **Task 1:**
 - Select a suitable case study related to cybersecurity. Then, provide the name of the case study you selected.
Tips:
 - Use a search engine like Google or an AI-powered tool like Perplexity to find potential case studies.
 - Ensure that the selected case study is relevant to key cybersecurity topics, such as cyberthreats and cyberattacks; networking and network security; database security; application security; AI; governance, risk, and compliance; penetration testing; threat hunting; encryption; cybersecurity architecture; incident response; digital forensics; and cryptography.
 - Write a clear and concise introduction to the case study that effectively sets the context for analysis and engages the reader. (Word limit: 150)

- **Task 2:**
 - Provide two credible citations that reference your selected case study.

Citation format examples are shown below:

1. Academic journal article:
Author(s). (Date). Title. *Journal title*, Volume(Issue), page numbers.

Yoo, C. S., & Lee, B. C. (2023). Optimizing cybersecurity risk in medical cyber-physical devices. *William & Mary Law Review*, 64(5), 1513–1554.

2. Webpage on a website:
Author(s). (Date). *Article title*. Website title. URL.

Doe, J. (2023, February 15). *Cybersecurity strategies: protecting your digital assets*. Cybersecurity Insights. <https://www.exampleofcitationforreference.com>.

- **Task 3:**
 - Provide your response against each section of the case study analysis template. Each section has pointers to help you systematically curate your responses. Also include references to support/justify your analysis.

Task 1: Find a suitable cybersecurity case study and write a concise introduction (in up to 150 words)

Task 1.1: Select a case study that should be relevant to the topics covered throughout the Cybersecurity Analyst courses, which could include:

1. Cyberthreats and cyberattacks
2. Networking and network security
3. Database security
4. Application security
5. Artificial intelligence (AI)
6. Governance, risk, and compliance
7. Penetration testing
8. Threat hunting
9. Encryption
10. Cybersecurity architecture
11. Incident response
12. Digital forensics
13. Cryptography

Look for studies that provide enough detail to analyze the successes, gaps, failures, and impact on the organization. Your chosen case study should involve real-world situations and scenarios. These studies will offer practical insights and lessons learned from actual experiences.

Task 1.2: Write a clear and concise introduction to the case study (**in not more than 150 words**) that effectively sets the context for analysis and engagement.

Submissions:

Task 1.1: Provide the name/title of the selected case study.

Task 1.2: Provide an introduction to the case study. (**in not more than 150 words**)

Task 2: Provide credible citations that reference your case study.

Find two credible citations that reference your selected case study. Your research should come from credible sources, such as academic journals, industry publications, reputable news sources, or other trusted organizations.

Submission: Two credible citations that reference your selected case study.

Task 3: Tabulate critical observations in the case study analysis template.

Use the given template to provide an analysis of the case study. The refresher overview of each section of the case study analysis template is shown below:

- **Task 3.1: Identify the root cause (Two root causes)**
 - Identify the initial point of failure, breach, or necessity for change.
 - Analyze the factors that contributed to the vulnerability or need for action.
- **Task 3.2: Document the actions taken (Two actions)**
 - List all steps and measures taken in response to the incident or to facilitate the change.
 - Include details such as dates, personnel involved, and resources used.
- **Task 3.3: Evaluate the effectiveness and timeliness (of the two actions taken in Task 3.2)**
 - Assess how effective and timely the actions were in addressing the incident or implementing the change.
 - Support your evaluation with metrics or feedback.
- **Task 3.4: Identify successes, gaps, and failures (One each)**
 - Identify the successes achieved, gaps found, and failures encountered during the response or change process.
 - Specific examples will help with clarity.
- **Task 3.5: Determine the impact on the organization (Two impacts)**
 - Discuss how the successes, gaps, and failures you have identified may affect the organization in the short and long term.
 - Consider both operational and strategic implications.
- **Task 3.6: Outline the lessons learned (Two lessons)**
 - Summarize the key lessons learned from the incident or change process.
 - Highlight both positive takeaways and areas for improvement.
- **Task 3.7: Recommendations for future actions (Two recommendations)**
 - Provide clear recommendations on what actions should be taken and what actions should be avoided in future incidents or change processes.
 - The recommendations should be actionable and based on the lessons learned.
- **Task 3.8: Draw a meaningful conclusion**
 - Reflect on the broader implications of the case study for the organization and the industry at large. Highlight any trends or patterns that emerged and their potential long-term effects.
 - Mention two references for your research.

Submissions:

Tasks 3.1 to 3.8: Provide your response against each section of the case study analysis template. Also, ensure to include references to support/justify your analysis wherever required.

Review criteria

A total of forty (40) points are allotted to the final project.

Your grade will be based on the following distribution of points:

Task 1: Find a suitable cybersecurity case study that you would like to evaluate and write a concise introduction (in up to 150 words) to set the context for analysis and engagement. (6 points)

Submissions required:

- Name of the chosen case study. (1 points)
- Introduction to the case study (in not more than 150 words). (5 points)

Task 2: Provide two credible citations that reference your selected case study. (2 points)

Submissions required: Two references (with proper citations) that give information on your selected case study. (2 points)

Task 3: Use the standard case study analysis template to document your observations. (32 points)

Submissions required: Provide your response against each section of the case study analysis template, with references to support/justify your analysis wherever required.
(Total eight [8] sub-tasks of four [4] points each)

Submitting your project

Review your submissions carefully to ensure they are complete and ready for evaluation before submitting your project.

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